

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY (RUET)

Department of Computer Science & Engineering

CSE 3206: Software Engineering Sessional

Lab #2: Software Process Models, Requirement Analysis & MVP Development

PROJECT DESIGN REPORT

PetCare: Pet Healthcare & Clinical Appointment Management System

Assigned Scenario: Group #37 • Suggested Model: Prototype Software Process Model

Team Information	Details
Team Name / Section	Group #07 from Section B (2nd 30)
Course Outcome	CO2: Design & implement software solutions addressing user requirements
Program Outcome	PO3: Design and Development of Software Solutions
Academic Session	2025–2026

Member	Assigned Role	Feature Branch	Technical Deliverables
Member 1	Architecture & Auth Lead	member1-feature	Express server, SQLite schema, User Auth (login/register), Navigation shell, Report 1-5, 10-12.
Member 2	Pet Profiles Lead	member2-feature	Pet CRUD, taxonomy, allergy tracker, vaccination log modal, Report 6-9 (12 FRs, 8 NFRs, 5 Use Cases).
Member 3	Appointments & Dashboard Lead	member3-feature	Appointment booking engine, status lifecycle, analytics metrics cards, Report 13-16 & PDF compilation.

3. Project Title

PetCare: An Interactive Pet Healthcare & Clinical Appointment Management System

4. Problem Statement

Pet owners face significant operational friction in maintaining longitudinal health records, tracking recurring vaccination schedules, and booking clinical veterinary appointments. Most domestic pet parents and veterinary clinics in developing regions rely on unindexed paper booklets, phone calls, or ad-hoc messaging. This disorganization causes critical vaccination lapses (e.g., missed rabies boosters), lost medical/allergy profiles, and frequent scheduling conflicts. A centralized, accessible, and intuitive digital system is urgently required to streamline clinical care.

5. Project Objectives

- **Centralized Health Records:** Digitize pet biometrics (species, breed, age, weight, microchip ID) and allergy notes into a single accessible portal.
- **Automated Vaccine Tracking:** Provide proactive status badges ('Up-to-Date', 'Due Soon', 'Overdue') to eliminate infectious disease risks.
- **Self-Service Clinical Booking:** Enable pet owners to schedule consultations, grooming sessions, and vaccinations with instant confirmation.
- **Role-Based Workspaces:** Deliver customized workflows for Pet Owners (focusing on their pets) and Veterinary Staff (managing clinic appointments).
- **Extensible Architecture:** Build an organized, modular foundation in Lab 2 to support Design Patterns and Testing in subsequent labs.

6. Stakeholder Analysis

Stakeholder	Primary Concerns	System Interactions
Pet Owners	Timely vaccine alerts, easy care booking, tracking pet weight/allergies.	Registers pets, views health cards, books clinical appointments.
Veterinarians	Quick access to pet allergies, medical histories, and daily appointment roster.	Reviews clinical history, records administered vaccines, completes visits.
Clinic Staff	Preventing double bookings, managing patient flow, walk-in registration.	Manages clinic appointment statuses, filters by date.
Course Evaluators	Process model justification, Git collaboration rigor, modularity, MVP feasibility.	Audits Git history, inspects architecture, tests live prototype in viva.

7. Functional Requirements (12 Specifications)

Req ID	Requirement Specification	Priority
FR-01	User authentication supporting registration, login, and session validation.	High
FR-02	Role switching between Pet Owner and Clinic Staff views.	High
FR-03	Pet profile creation with Name, Species, Breed, Age, Gender, Weight, and Microchip ID.	High
FR-04	Pet profile modification and cascading record deletion.	High
FR-05	Pet search by name/breed and taxonomic filtering by species (Dog, Cat, Rabbit, Other).	Medium
FR-06	Vaccination logging with administered date, next due date, and automated overdue flags.	High
FR-07	Allergy and medical observations tracking with visual health alerts.	Medium
FR-08	Appointment scheduling specifying pet, service type, date, time slot, and specialist.	High
FR-09	Appointment lifecycle state transitions (Scheduled, In-Progress, Completed, Cancelled).	High
FR-10	Appointment status tab filtering (All, Scheduled, Completed, Cancelled).	Medium
FR-11	Dashboard analytical metric counters (Total Pets, Upcoming Visits, Overdue Vaccines).	High
FR-12	Sample dataset reset trigger for laboratory evaluation demonstration.	Low

8. Non-Functional Requirements (8 Specifications)

Req ID	Attribute	Specification Criteria
NFR-01	Performance	Local API queries and database operations respond within 300 milliseconds.
NFR-02	Usability	Responsive layout using CSS Grid/Flexbox for seamless desktop and tablet viewing.
NFR-03	Modularity	Strict separation between Presentation, API Routing, and Database layers.
NFR-04	Portability	Zero-dependency SQLite storage requiring no external database server installations.
NFR-05	Data Integrity	Foreign key constraints with ON DELETE CASCADE prevent orphan records.
NFR-06	Security	Input sanitization and payload validation to prevent injection or malformed data.
NFR-07	Reliability	Structured HTTP status codes and user-friendly error feedback modals.
NFR-08	Traceability	All deliverables version-controlled across 3 dedicated feature branches and PRs.

9. Five Key User Stories & Use Cases

UC-01: Registering a Pet Profile

Primary Actor: Domestic Pet Owner • **Precondition:** Pet Profile Form filled with biometrics & chip ID.

Postcondition: Pet record persisted, card displayed in grid, dashboard counters incremented.

UC-02: Booking a Clinical Service

Primary Actor: Domestic Pet Owner • **Precondition:** Pet selected, service chosen, date & slot confirmed.

Postcondition: Appointment registered as 'Scheduled' and visible in clinic roster.

UC-03: Reviewing Overdue Health Alerts

Primary Actor: Pet Owner / Vet • **Precondition:** User opens dashboard or pet list.

Postcondition: System flags overdue vaccines with red warning badges and recommendation.

UC-04: Completing a Clinical Service

Primary Actor: Attending Veterinarian • **Precondition:** Doctor finishes treatment and clicks 'Complete'.

Postcondition: Status transitions to 'Completed' and dashboard analytics update in real time.

UC-05: Logging a Vaccine Administration

Primary Actor: Veterinary Specialist • **Precondition:** Attending doctor submits vaccine name, dose date, next due date.

Postcondition: New record added to vaccination table; pet's alert badge clears to 'Up-to-Date'.

10. Selected Software Process Model: Prototype Model

The **Prototype Software Process Model** was chosen as the most suitable methodology. In this model, an initial functional prototype demonstrating core requirements is constructed rapidly, evaluated by stakeholders, and iteratively refined based on real-world feedback.

11. Justification of the Prototype Model

- **Qualitative & Evolving User Requirements:** Pet healthcare involves non-technical users whose operational needs (e.g. appointment steps, allergy visibility) are best understood by interacting with a working system.
- **Early Usability Validation:** Veterinary staff can test whether emergency appointment bookings and health status tags are sufficiently intuitive prior to investing in complex enterprise modules.
- **Solid Architecture for Semester Labs:** Building this functional MVP in Lab 2 establishes clean domain entities and endpoints that will seamlessly host Design Patterns (Lab 3/4) and automated test suites (Lab 5).

12. Comparison with Alternative Models

Evaluation Metric	Prototype Model (Chosen)	Waterfall Model	Spiral Model
Requirement Flexibility	High (Refined via working mockups)	Low (Frozen at analysis)	Moderate (Refined per cycle)
Stakeholder Feedback	Immediate working feedback in Lab 2	Late (At the end of development)	Periodic after risk evaluations
Suitability for MVP	Optimal for small team & rapid delivery	Poor (High risk of rework)	Poor (Excessive administrative overhead)

13. MVP Design Overview

Architecture: 3-Tier Layered Design comprising a Vanilla JS SPA frontend, an Express.js REST API router layer, and an embedded relational SQLite database engine with foreign key enforcement and JSON fallback. **Core Modules:** (1) Authentication & Role Switching, (2) Pet Profiles & Medical Card CRUD, (3) Clinical Appointment Scheduling & Analytics Dashboard.

14. GitHub Collaboration Evidence

The team adhered strictly to the mandated GitHub workflow: main branch protected as production baseline, with 3 dedicated feature branches (member1-feature, member2-feature, member3-feature) authored independently and integrated into main via reviewed Pull Requests.

15. Challenges Encountered & Solutions

- **Challenge 1 (Native SQLite Portability):** Native C++ SQLite binaries often fail on machines lacking build tools. *Solution:* Used Node 22 built-in node:sqlite with a zero-dependency JSON fallback engine.
- **Challenge 2 (Express 5 Route Matching):** Express 5 deprecated the catch-all asterisk route syntax. *Solution:* Implemented standard middleware notation (app.use((req, res) => ...)).
- **Challenge 3 (Multi-Role Viva Demonstration):** Switching accounts rapidly during live presentation. *Solution:* Created one-click quick-login buttons for Sarah (Pet Owner) and Dr. Rahman (Veterinarian).

16. Conclusion

The Pet Care Management System MVP fulfills all criteria of RUET CSE 3206 Lab 2. The team demonstrated professional software engineering practices, clear work division, thorough requirement analysis, and a functional, modular prototype ready for semester-long enhancement.